Motions Of Charged Particles In Static Electromagnetic Field

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Let us consider that, in the space, the electric field is $\mathbf{E} = E_0\hat{a_y}$ and the magnetic flux density is $\mathbf{B} = B_0\hat{a_z}$. A particle with mass m have an initial velocity $v_x\hat{a_x} + v_y\hat{a_y} + v_z\hat{a_z}$.

1 Analyze the Motions with Basic Lorentz force

We know that

$$\mathbf{F} = qv_yB_0\hat{a_x} + (qE_0 - qv_xB_0)\hat{a_x}.$$

Therefore, we have

$$\begin{aligned} m\frac{dv_x}{dt} &= qv_yB_0 \\ m\frac{dv_y}{dt} &= qE_0 - qv_xB_0 \\ m\frac{dv_z}{dt} &= 0. \end{aligned}$$

Let us only consider the motion on xy plane, since v_z is irrelevant:

$$\frac{dv_y^2}{dt^2} = \frac{-qB_0}{m}v_y$$

We have

$$\begin{aligned} v_y &= V\cos(\frac{qB_0}{m}t + \phi) \\ v_x &= \frac{E_0}{B_0} + V\sin(\frac{qB_0}{m}t + \phi) \end{aligned}$$

Let us take a look at the trajectories. We let $E_0 = B_0 = m = 1, \phi = \pi/8$.

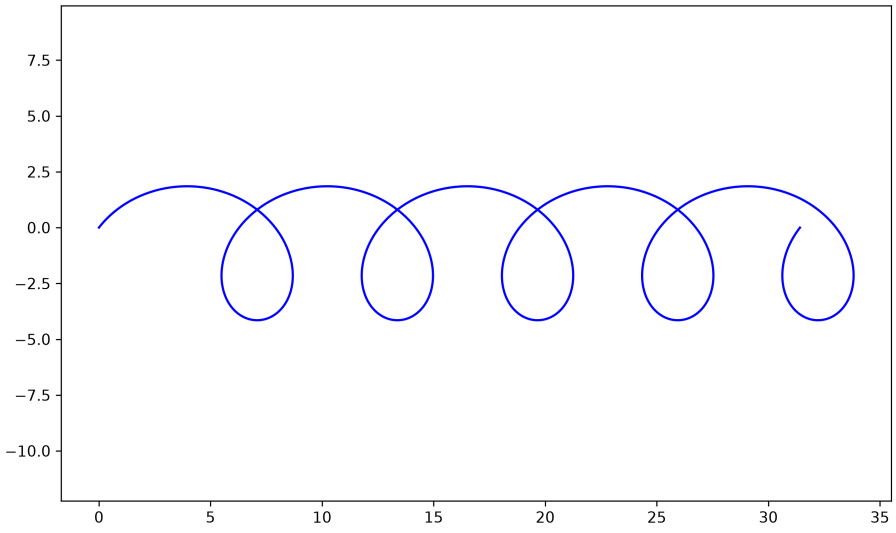


Figure 1: $V = 3, q = +1$

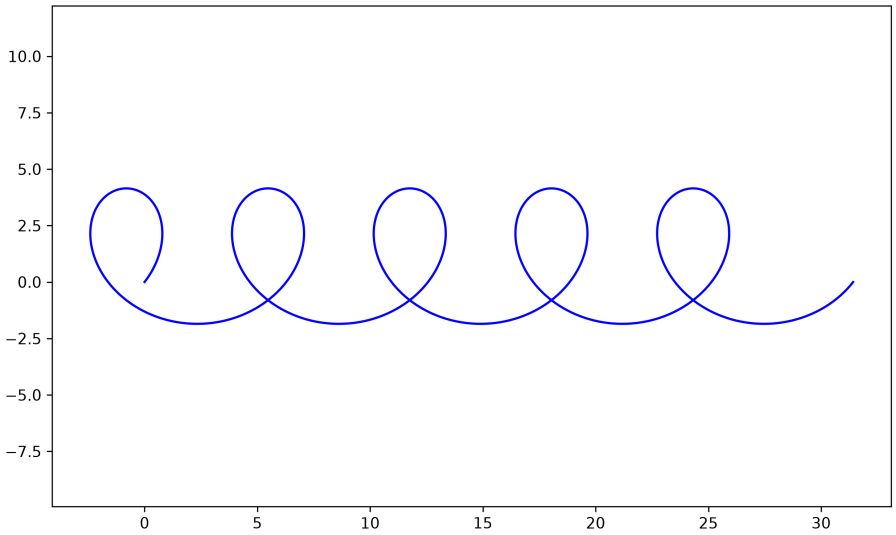


Figure 2: $V = 3, q = -1$

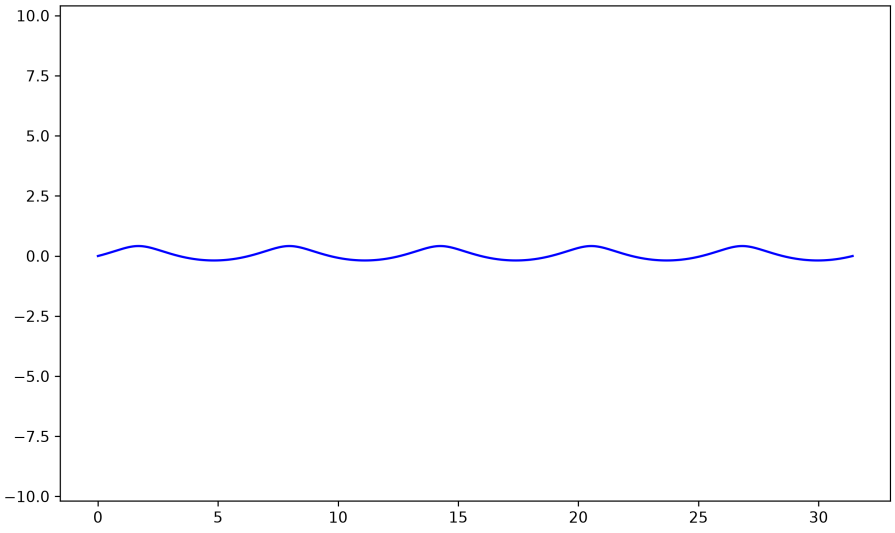


Figure 3: $V = 0.3, q = -1$

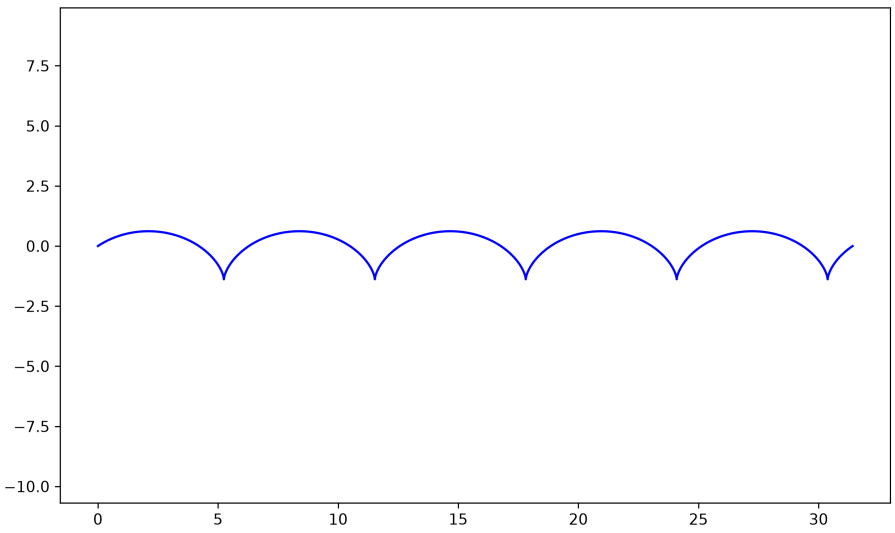


Figure 4: $V = 1, q = +1$

2 Average Velocity

Easily we see that the average velocity is $\frac{E_0}{B_0}\hat{a_x}$. However, Let us look at the Galilean transformation formula for fields, that is $\mathbf{E}' = \mathbf{E} + \mathbf{v} \times \mathbf{B}$. Observe that if $\mathbf{v} = \frac{E_0}{B_0}\hat{a_x}$, the new electric field is zero. In the coordinate system, with the magnetic flux density only, we see the particle rotates at the same place!

There *is* something weird without the assumptions of the Spacial Theory of Relativity. But, let us just take a sip of our tea and get some rest.